Prachi Sanaliya
AU2340035

CSC_360
Class Notes

13/8/26

Topics Learned:
- Setup for Java programming and Maven.
- Setup of the environment required for Graphics Programming in Java.
- Setup of GitHub Repository using Git and Command Prompt.
- Creation and running a basic Java Program which creates Geometric square.

Java Setup and Validation:
- Installation and validation of the Java Development Kit (JDK).
- Validation of java and Java Compiler was performed by commands:
  ~ java -version
  ~ javac -version
  ~ javac – Java compiler.
  ~ java – Used to run compiled Java programs.

Maven Setup:
- Installation and Setup of Apache Maven.
- Apache Maven is used for building and managing Java Projects.
- Installation and validation of Apache Maven was done using:
  ~ mvn -version
- Apache Maven Setup was associated with the Java JDK.

Setting Up the GitHub Repository:
- Used an already existing GitHub repository.

- Cloning of the repository on the local machine was done through the following command:
  ~ git clone <repository-url>
- Accessing the cloned repository was done via the local machine via the Command Prompt interface.
- The file created in the class exercise was saved in the cloned repository.

Java Graphics Application:
- A simple Java program that draws a square through lines was created.
- Java Swing and AWT Graphics were used to create the graphics window and the square.
- JFrame was used to create the graphics window for the application.
- JPanel was used as the drawing canvas.
- Graphics2D was used in drawing the graphic objects.
- drawRect() method was used in drawing the square.
- BasicStroke was used in creating a solid and unbroken line for the square.

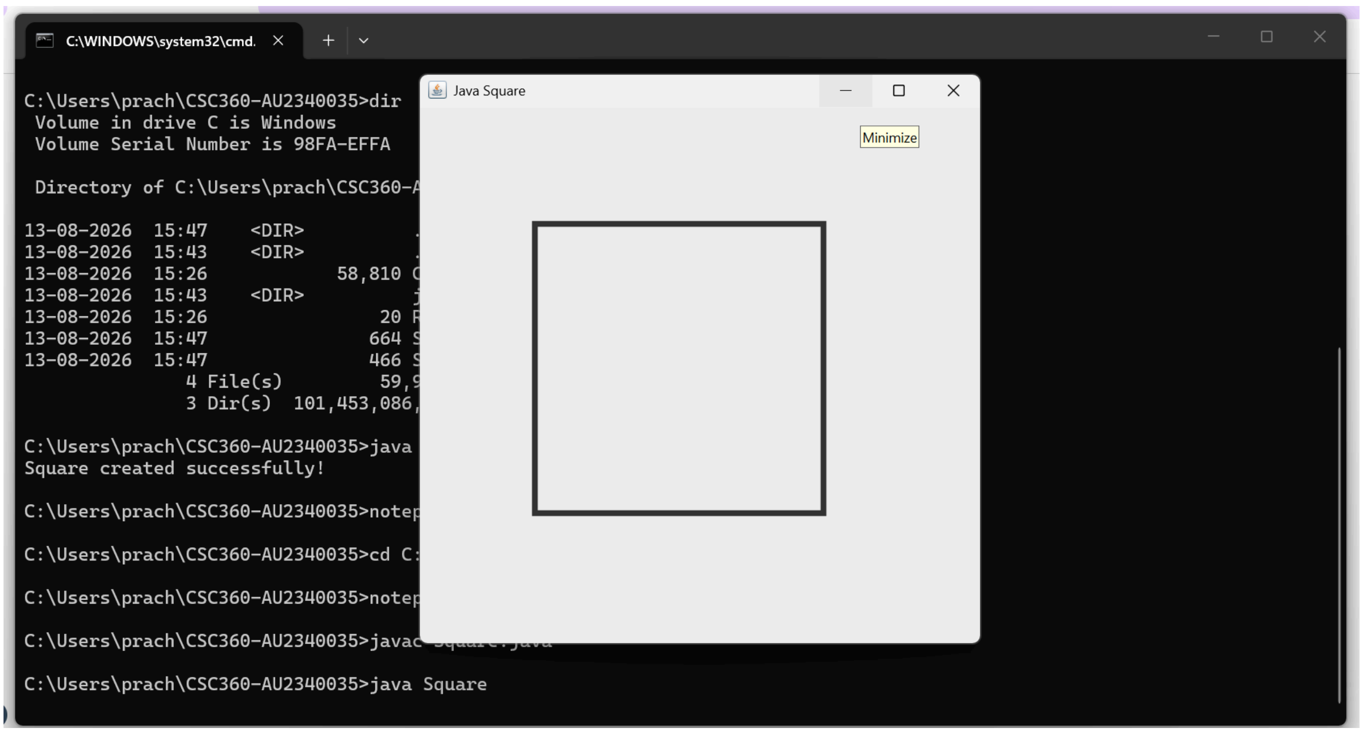
Compilation and Execution of the Program:

- Java source code was written as: Square.java
- The compilation of the program was done using: javac Square.java
- Execution of the program was done using: java Square
- Running the program opened a window displaying the square in graphical form.

Git Commands Used:

- After writing the Java program and ensuring that it runs, the files were then committed into the Git repository.
- Checking status of repository: git status

- Adding files: git add .
- Setting up Git identity if required: git config --global user.name ,git config --global user.email
- Committing changes: git commit -m "Add Java square program"
- Pushing to GitHub: git push origin main



PrachiSanaliya Add Java square program

Code | Blame | 37 lines (26 loc) · 862 Bytes

```java
import javax.swing.*;
import java.awt.*;

public class Square extends JPanel {

    @Override
    protected void paintComponent(Graphics g) {
        super.paintComponent(g);

        Graphics2D g2 = (Graphics2D) g;

        // Make the lines smooth
        g2.setRenderingHint(
            RenderingHints.KEY_ANTIALIASING,
            RenderingHints.VALUE_ANTIALIAS_ON
        );

        // Solid line with thickness 5 pixels
        g2.setStroke(new BasicStroke(5));

        // Draw a complete square
        g2.drawRect(100, 100, 250, 250);
    }

    public static void main(String[] args) {

        JFrame frame = new JFrame("Java Square");

        frame.add(new Square());

        frame.setSize(500, 500);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setLocationRelativeTo(null);

        frame.setVisible(true);
    }
}
```